Week 6-8: Free Boundary Problems & Application to American Options

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1 American Options as Free Boundary Problems

Starting from a European put option, whose price $P(S,t)$ satisfies the BS equation:

$$\frac{\partial P}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 P}{\partial S^2} + rS\frac{\partial P}{\partial S} - rP = 0. \tag{1}$$

For a European put, the terminal condition is $P(S,T) = \max(E - S, 0)$, together with the boundary conditions

$$P(0,t) = Ee^{-r(T-t)}; P(S,t) \to 0 \quad \text{as } S \to \infty. \tag{2}$$

In contrast, an American put can be exercised at any time prior to maturity, which introduces the constraint

$$P(S,t) \geq max(E - S, 0), \tag{3}$$

otherwise, people can buy the option and exercise immediately, causing the arbitrage opportunities. This inequality fundamentally changes the structure of the problem. At each point (S,t), there are two possible regimes:

- Continuation region: If $P(S,t) > E - S$, it is optimal to continue holding the option, where the price satisfies the BS equation.
- Exercise region: If $P(S,t) = E - S$, it is optimal to exercise immediately, where the option value coincides with its payoff.

Therefore, there exist the optimal exercise price $S_f(t)$ to divide the continuation and the exercise region. Formally,

$$\begin{cases} P(S,t) = E - S, & S \leq S_f(t), \\ P(S,t) > E - S, & S > S_f(t). \end{cases} \tag{4}$$

At this boundary, the solution must satisfy two matching conditions:

$$\begin{aligned} P(S_f(t),t) &= E - S_f(t) \quad \text{(value matching)}, \\ \frac{\partial P}{\partial S}(S_f(t),t) &= -1 \quad \text{(smooth pasting)}. \end{aligned} \tag{5}$$

Since $S_f(t)$ is not a *priori* known function, thus determining it forms the free boundary problem, where both the function $P(S,t)$ and the boundary $S_f(t)$ are unknown and must be solved simultaneously. The American put option as an free boundary problem is summarized in Figure 1.

2 Methods for Free Boundary Problems

There are two strategies to solve the free boundary problems.

1. Boundary tracking: track the free boundary as part of the time-stepping process. But given that equations in (5) are both implicit, we need to use the implicit free boundary methods which are not our focus.

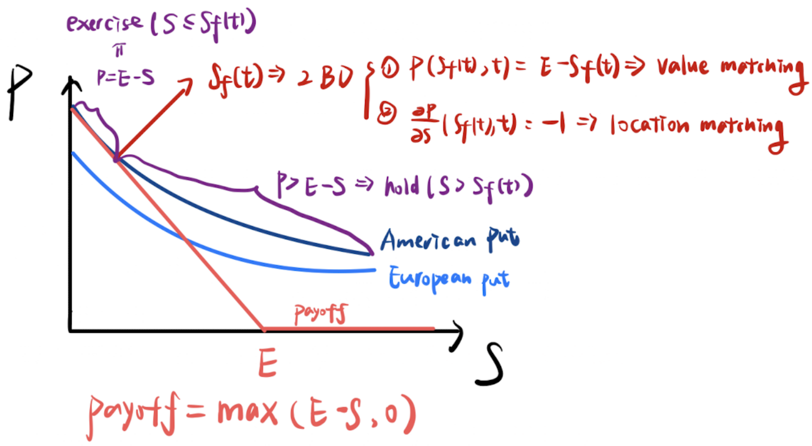

Figure 1: Free Boundary Illustration for American Put Option

2. Transformation: Transformation reduce the problems into a fixed boundary, from which the free boundary can be inferred afterwards. Two main transformation methods:

    (a) Linear complementarity problems (LCP): This is our main focus transformation, and after the transformation, the following section shows how we use the FDM scheme to discretize it and use the projected successive over relaxation (PSOR) method to solve it. This LCP $\Rightarrow$ FDM $\Rightarrow$ PSOR scheme is applied to both obstacle problems and American put options correspondingly.

    (b) Variational inequalities (VI): Since this prepares a weak formulation, finite element method is more useful to solve it. But this week only focuses on the construction of VI for both obstacle problem and American option.

2.1 Transformation 1: LCP

2.1.1 Obstacle Problem

Model Problem Obstacle problem is the a canonical free boundary problem with clear physical and mathematical interpretations. A typical example is illustrated in Figure 2. Formally, the obstacle

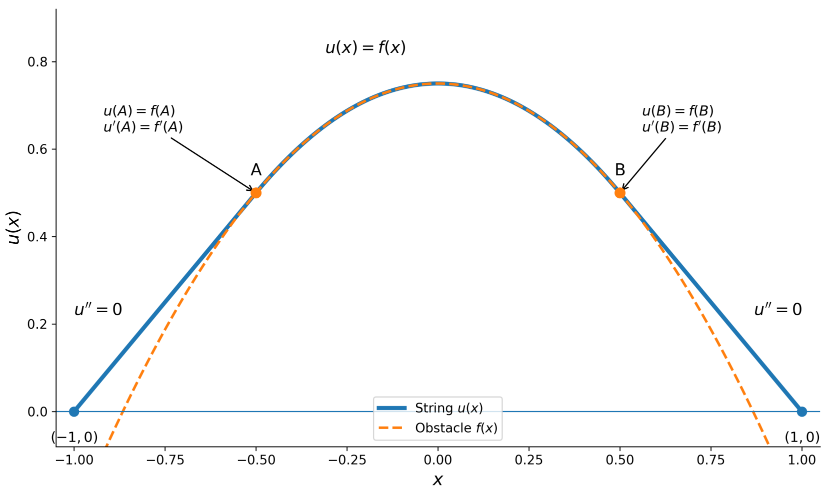

Figure 2: Illustration for A Obstacle Problem

problem consists in finding a function $u(x)$ and two unknown points A and B such that

$$\begin{cases} u''(x) = 0, & -1 < x < A, \\ u(x) = f(x), & A \le x \le B, \\ u''(x) = 0, & B < x < 1, \\ u(-1) = 0, \quad u(1) = 0, & \\ u(A) = f(A), \quad u'(A) = f'(A), & \\ u(B) = f(B), \quad u'(B) = f'(B). & \end{cases} \tag{6}$$

Interpretation:

- Function: The function $u(x)$ represents the vertical displacement of the string, while $f(x)$ defines the shape of the obstacle. This problem describes the equilibrium shape of an elastic string that is fixed at both ends and constrained to lie above a given obstacle $f(x)$.
- $u''(x) = 0$: In regions where the string does not touch the obstacle, namely $(-1, A)$ and $(B, 1)$, the equation indicates that the string is straight, reflecting the absence of external forces other than tension.
- $u(x) = f(x)$: On the interval $[A, B]$, the string is in contact with the obstacle and therefore coincides with it.
- Transition points A and B: At these points, both the displacement and the slope are continuous, as expressed by the conditions $u(A) = f(A)$, $u'(A) = f'(A)$ and $u(B) = f(B)$, $u'(B) = f'(B)$. These conditions ensure that the string meets the obstacle smoothly without forming kinks, reflecting a balance of forces at the contact points.

Thus, the obstacle problem is characterized by the fact that both the solution $u(x)$ and the contact region $[A, B]$ are unknown and must be determined simultaneously.

LCP Format This problem (6) can be then formulated into LCP form:

$$u''(u - f) = 0, \quad -u'' \ge 0, \quad (u - f) \ge 0, \tag{7}$$

subject to the conditions:

$$u(-1) = u(1) = 0, \qquad u,\, u' \text{ are continuous.} \tag{8}$$

This means that, the obstacle is active ($u = f$) or the PDE holds ($u'' = 0$), but not both.

FDM Scheme The FDM is then applied to discretize this problem, using a uniform grid

$$x_n = n\,\delta_x, \quad -N \le n \le N, \quad \delta_x = \frac{1}{N}, \tag{9}$$

so that $x_{-N} = -1$ and $x_N = 1$. Using central differences, the second derivative is approximated by

$$u''(x_n) \approx \frac{\tilde{u}_{n+1} - 2\tilde{u}_n + \tilde{u}_{n-1}}{(\delta_x)^2}, \tag{10}$$

where $\tilde{u}_n$ denotes the numerical approximation of $u(x_n)$. We also define $f_n = f(x_n)$. The boundary conditions become

$$\tilde{u}_{-N} = \tilde{u}_N = 0. \tag{11}$$

For interior nodes, the continuous LCP is transformed into the discrete system

$$\mathbf{B}\tilde{\mathbf{u}} \ge 0, \quad \tilde{\mathbf{u}} \ge \mathbf{f}, \quad (\tilde{\mathbf{u}} - \mathbf{f})^\top (\mathbf{B}\tilde{\mathbf{u}}) = 0. \tag{12}$$

The matrix $\mathbf{B}$ is the discrete Laplacian

$$\mathbf{B} = \begin{pmatrix} 2 & -1 & 0 & \cdots & 0 \\ -1 & 2 & -1 & \cdots & 0 \\ 0 & -1 & 2 & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & -1 \\ 0 & \cdots & 0 & -1 & 2 \end{pmatrix}, \tag{13}$$

and

$$\tilde{\mathbf{u}} = \begin{pmatrix} \tilde{u}_{-N+1} \\ \vdots \\ \tilde{u}_{N-1} \end{pmatrix}, \qquad \mathbf{f} = \begin{pmatrix} f_{-N+1} \\ \vdots \\ f_{N-1} \end{pmatrix}. \tag{14}$$

PSOR Framework The system (12) can be solved via PSOR method iteratively. But before the application to this simple problem, we need a general PSOR framework. We consider the general LCP problem

$$\mathbf{A}\mathbf{x} \geq \mathbf{b}, \quad \mathbf{x} \geq \mathbf{c}, \quad (\mathbf{x} - \mathbf{c})^\top (\mathbf{A}\mathbf{x} - \mathbf{b}) = 0, \tag{15}$$

where $\mathbf{A}$ is symmetric positive definite. The PSOR method is an iterative scheme. Starting from an initial guess $\mathbf{x}^0 \geq \mathbf{c}$, we update each component sequentially. For the i-th row of $\mathbf{A}\mathbf{x} = \mathbf{b}$,

$$A_{i1}x_1 + \cdots + A_{ii}x_i + \cdots + A_{in}x_n = b_i, \tag{16}$$

we isolate x_i:

$$x_i = \frac{1}{A_{ii}} \left(b_i - \sum_{j<i} A_{ij}x_j - \sum_{j>i} A_{ij}x_j \right). \tag{17}$$

However, at iteration $k+1$, we use the Gauss–Seidel splitting, i.e., updated values for $j < i$ (before i, updated already) and the old values (not yet updated) for $j > i$, leading to the intermediate value as a candidate solution

$$y_i^{k+1} = \frac{1}{A_{ii}} \left(b_i - \sum_{j=1}^{i-1} A_{ij}x_j^{k+1} - \sum_{j=i+1}^{n} A_{ij}x_j^{k} \right). \tag{18}$$

The projected SOR update is then

$$x_i^{k+1} = \max\left(c_i,\ x_i^k + \omega\big(y_i^{k+1} - x_i^k\big)\right), \tag{19}$$

where $\omega \in (0, 2)$ is the relaxation parameter, which controls the speed of convergence. The iteration proceeds sequentially for $i = 1, \dots, n$. We stop when

$$\|\mathbf{x}^{k+1} - \mathbf{x}^k\| < \varepsilon, \tag{20}$$

and take $\mathbf{x}^{k+1}$ as the numerical solution. The whole procedure is summarized in Algorithm 1. This

Algorithm 1 Framework of the PSOR method

Require: Mesh, boundary conditions, initial guess $\mathbf{x}^0 \geq \mathbf{c}$, relaxation parameter $\omega \in (0, 2)$, tolerance $\varepsilon > 0$
Ensure: Approximate solution $\mathbf{x}$

```
Construct the computational mesh and identify all interior nodes.
Impose the boundary conditions.
Initialize x^0 satisfying the constraint x^0 ≥ c.
Set k = 0.
repeat
    Update the leftmost interior node (adjust for boundary).
    for i = 2, ..., n − 2 do
        Compute y_i^{k+1} = (1/A_ii) (b_i − Σ_{j=1}^{i−1} A_ij x_j^{k+1} − Σ_{j=i+1}^{n} A_ij x_j^k)
        Update x_i^{k+1} = max(c_i, x_i^k + ω(y_i^{k+1} − x_i^k))
    end for
    Update the rightmost interior node (adjust for boundary).
    Compute the error err = max_i |x_i^{k+1} − x_i^k|.
    Set k ← k + 1.
until err < ε
Return x^k.
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framework is then applied to the above problem (12), following the steps in Algorithm 2.

Algorithm 2 PSOR for the discrete obstacle problem

Require: Integer N, obstacle values $\{f_n\}_{n=-N}^{N}$, relaxation parameter $\omega \in (0,2)$, tolerance $\varepsilon > 0$
Ensure: Approximate discrete solution $\{v_n\}_{n=-N}^{N}$

Construct the uniform mesh $x_n = n\delta_x$, for $-N \le n \le N$, with $\delta_x = \frac{1}{N}$.
Impose the boundary conditions $v_{-N} = 0$ and $v_N = 0$.
Choose an initial guess $\{v_n^0\}$ such that $v_n^0 \ge f_n$ for all interior nodes $-N+1 \le n \le N-1$.
Set $k = 0$.
repeat
  Update the **leftmost** interior node $n = -N+1$:
  Compute $y_{-N+1}^{k+1} = \frac{1}{2}\left(v_{-N}^{k+1} + v_{-N+2}^{k}\right)$
  Update $v_{-N+1}^{k+1} = \max\left(f_{-N+1},\ v_{-N+1}^{k} + \omega\big(y_{-N+1}^{k+1} - v_{-N+1}^{k}\big)\right)$
  for $n = -N+2, \ldots, N-2$ **do**
    Compute $y_n^{k+1} = \frac{1}{2}\left(v_{n-1}^{k+1} + v_{n+1}^{k}\right)$
    Update $v_n^{k+1} = \max\left(f_n,\ v_n^k + \omega\big(y_n^{k+1} - v_n^k\big)\right)$
  end for
  Update the **rightmost** interior node $n = N-1$:
  Compute $y_{N-1}^{k+1} = \frac{1}{2}\left(v_{N-2}^{k+1} + v_{N}^{k+1}\right)$
  Update $v_{N-1}^{k+1} = \max\left(f_{N-1},\ v_{N-1}^{k} + \omega\big(y_{N-1}^{k+1} - v_{N-1}^{k}\big)\right)$
  Compute the error $err = \max_{-N+1 \le n \le N-1} |v_n^{k+1} - v_n^k|$.
  Set $k \leftarrow k+1$.
until $err < \varepsilon$
Return $\{v_n^k\}_{n=-N}^{N}$.

Numerical Example: 1D We first consider the one-dimensional obstacle problem on the domain $[-1,1]$ in equation (7), and (8). The obstacle function is chosen as

$$f(x) = \max\left(0.35e^{-\left(\frac{x+0.55}{0.18}\right)^2} + 0.85e^{-\left(\frac{x-0.20}{0.23}\right)^2} - 0.02,\ 0\right), \tag{21}$$

which produces a two-hump structure. Parameter setting:

- Grid size: $N = 200$ (total points 401)
- Relaxation parameter: $\omega = 1.6$
- Tolerance: $\varepsilon = 10^{-10}$
- Initial guess: $v^0 = \max(f, 0)$

Results are recorded in Table 1 and Figure 3.

Table 1: PSOR performance for 1D Obstacle

Quantity	Value
Iterations	10694
Final $\lVert \mathbf{v}^{k+1} - \mathbf{v}^k \rVert_2$	9.99×10^{-11}
$\min(v - f)$	0.0
$\min(Bv)$	-2.09×10^{-12}
$\max\lvert(v-f)\cdot Bv\rvert$	1.06×10^{-12}

Numerical Example: 2D The method can also be applied to 2D on the square domain $\Omega = [-1,1] \times [-1,1]$:

$$-\Delta u \ge 0, \qquad u \ge f, \qquad (u-f)(-\Delta u) = 0 \quad \text{in } \Omega, \tag{22}$$

subject to the homogeneous Dirichlet boundary condition

$$u = 0 \qquad \text{on } \partial\Omega. \tag{23}$$

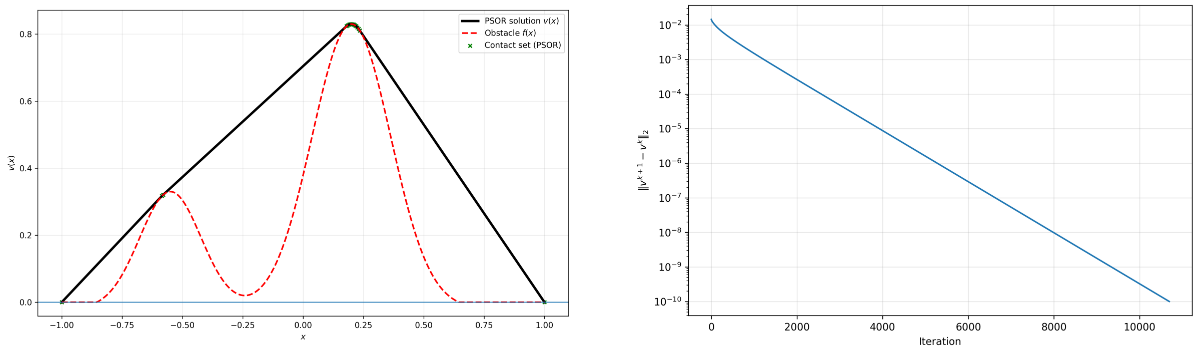

Figure 3: Numerical solution of the obstacle problem and convergence behavior of the PSOR method.

The obstacle is chosen as a smooth two-hump function

$$f(x,y) = \max\Big(0.45 \exp\Big(-\Big(\frac{x+0.45}{0.22}\Big)^2 - \Big(\frac{y+0.10}{0.28}\Big)^2\Big) \\ + 0.85 \exp\Big(-\Big(\frac{x-0.20}{0.26}\Big)^2 - \Big(\frac{y-0.15}{0.22}\Big)^2\Big) \\ - 0.02,\ 0\Big). \tag{24}$$

The domain is then discretized by a uniform square grid with mesh sizes $h_x = h_y = h$, and the continuous problem is approximated by the discrete complementarity system in (12), but here $\mathbf{B}$ is the matrix associated with the standard five-point discretization of $-\Delta$. The resulting discrete system is solved by the PSOR method described in Algorithm 1. Parameter setting:

- Grid size: 81×81
- Domain: $[-1, 1] \times [-1, 1]$
- Relaxation parameter: $\omega = 1.7$
- Tolerance: $\varepsilon = 10^{-8}$
- Maximum iterations: 30000
- Initial guess: obstacle-based initialization

The results are recorded in Table 2. For the Figure 4, from left to right: the PSOR solution and obstacle surface, the top-view contour of the solution with obstacle and contact boundary, and the convergence history of the PSOR iteration measured by $\|u^{k+1} - u^k\|_2$.

Table 2: PSOR performance for 2D Obstacle

Metric	Value
Iterations	857
Final $\|\mathbf{u}^{k+1} - \mathbf{u}^k\|_2$	9.85×10^{-9}
$\min(u - f)$	0.0
$\min(-\Delta_h u)$	-1.91×10^{-7}
$\max\|(u - f)(-\Delta_h u)\|$	4.53×10^{-8}

2.1.2 American options

Model Problem Different like the obstacle problem which only consider the spatial variable, the American option problem consider the discretization of both space (price) and time variable. To generalize it, we first transform it into the diffusion type equation. For both the American put and call, the governing equation in the continuation region is

$$\frac{\partial u}{\partial \tau} = \frac{\partial^2 u}{\partial x^2}. \tag{25}$$

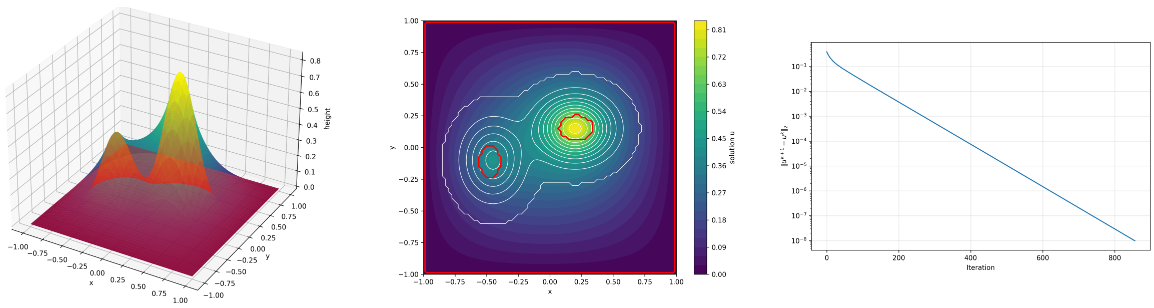

Figure 4: Numerical results for the 2D obstacle problem.

For American put:

$$\begin{aligned} u(x,0) &= \max\left(e^{\frac{1}{2}(k_2-1)x} - e^{\frac{1}{2}(k_2+1)x},\, 0\right), \\ u(x,\tau) &\geq e^{\frac{1}{4}\left((k_2-1)^2+4k_1\right)\tau} \max\left(e^{\frac{1}{2}(k_2-1)x} - e^{\frac{1}{2}(k_2+1)x},\, 0\right), \\ \lim_{x\to\infty} u(x,\tau) &= 0. \end{aligned} \tag{26}$$

For American call:

$$\begin{aligned} u(x,0) &= \max\left(e^{\frac{1}{2}(k_2+1)x} - e^{\frac{1}{2}(k_2-1)x},\, 0\right), \\ u(x,\tau) &\geq e^{\frac{1}{4}\left((k_2-1)^2+4k_1\right)\tau} \max\left(e^{\frac{1}{2}(k_2+1)x} - e^{\frac{1}{2}(k_2-1)x},\, 0\right), \\ \lim_{x\to-\infty} u(x,\tau) &= 0. \end{aligned} \tag{27}$$

The key transformed variables are

$$x = \ln\left(\frac{S}{E}\right), \qquad \tau = \frac{\sigma^2}{2}(T-t), \tag{28}$$

and the dimensionless parameters are

$$k_1 = \frac{2r}{\sigma^2}, \qquad k_2 = \frac{2(r-q)}{\sigma^2}, \tag{29}$$

where q stands for the dividends, without q, $k_1 = k_2$ and American call option is the same as European call option.

LCP Format Similar to the obstacle problem, all the American option valuation problems can be written in the compact LCP form:

$$\left(\frac{\partial u}{\partial \tau} - \frac{\partial^2 u}{\partial x^2}\right) \geq 0, \qquad u(x,\tau) - g(x,\tau) \geq 0, \tag{30}$$

$$\left(\frac{\partial u}{\partial \tau} - \frac{\partial^2 u}{\partial x^2}\right)\big(u(x,\tau) - g(x,\tau)\big) = 0. \tag{31}$$

Here $g(x,\tau)$ is the transformed payoff constraint function. For American put:

$$g(x,\tau) = e^{\frac{1}{4}\left((k_2-1)^2+4k_1\right)\tau} \max\left(e^{\frac{1}{2}(k_2-1)x} - e^{\frac{1}{2}(k_2+1)x},\, 0\right). \tag{32}$$

For American call:

$$g(x,\tau) = e^{\frac{1}{4}\left((k_2-1)^2+4k_1\right)\tau} \max\left(e^{\frac{1}{2}(k_2+1)x} - e^{\frac{1}{2}(k_2-1)x},\, 0\right). \tag{33}$$

FDM Scheme Then, the FDM scheme is applied to American put option only. In contrast to the stationary obstacle problem, where only the spatial variable is discretized, the American option problem is time-dependent. As a result, the discretization produces not a single complementarity system, but a sequence of linear complementarity problems, one at each time level. We first construct a uniform mesh in the (x, τ)-plane,

$$x_n = n\,\delta_x, \qquad \tau_m = m\,\delta_\tau, \tag{34}$$

and write

$$u_n^m = u(x_n, \tau_m), \qquad g_n^m = g(x_n, \tau_m). \tag{35}$$

Here u_n^m denotes the numerical approximation to the transformed option value, while g_n^m is the transformed payoff constraint evaluated at the same grid point.

Parabolic PDE We begin with the time derivative. Since the problem evolves forward in the transformed time variable τ, we approximate

$$\frac{\partial u}{\partial \tau}(x_n, \tau_m) \approx \frac{u_n^{m+1} - u_n^m}{\delta_\tau}. \tag{36}$$

As in the obstacle problem, the spatial operator is approximated by the standard central difference formula. To keep the scheme general, we use the θ-method for a general framework.

$$\frac{\partial^2 u}{\partial x^2}(x_n, \tau_m) \approx \theta\frac{u_{n+1}^{m+1} - 2u_n^{m+1} + u_{n-1}^{m+1}}{(\delta_x)^2} + (1-\theta)\frac{u_{n+1}^m - 2u_n^m + u_{n-1}^m}{(\delta_x)^2}. \tag{37}$$

The parameter $\theta \in [0, 1]$ determines how the diffusion term is evaluated in time:

- $\theta = 0$: explicit method
- $\theta = \frac{1}{2}$: Crank–Nicolson method
- $\theta = 1$: implicit method

Substituting the approximations for the time derivative and spatial derivative into the operator $\frac{\partial u}{\partial \tau} - \frac{\partial^2 u}{\partial x^2}$, we obtain the discrete inequality

$$\frac{u_n^{m+1} - u_n^m}{\delta_\tau} - \theta\frac{u_{n+1}^{m+1} - 2u_n^{m+1} + u_{n-1}^{m+1}}{(\delta_x)^2} - (1-\theta)\frac{u_{n+1}^m - 2u_n^m + u_{n-1}^m}{(\delta_x)^2} \geq 0. \tag{38}$$

Introducing the mesh ratio

$$\alpha = \frac{\delta_\tau}{(\delta_x)^2}, \tag{39}$$

this can be rewritten as

$$u_n^{m+1} - \alpha\theta\big(u_{n+1}^{m+1} - 2u_n^{m+1} + u_{n-1}^{m+1}\big) \geq u_n^m + \alpha(1-\theta)\big(u_{n+1}^m - 2u_n^m + u_{n-1}^m\big). \tag{40}$$

To make it a standard and compact FDM format, we then define the RHS to be b_n^m, so that the inequality becomes

$$u_n^{m+1} - \alpha\theta\big(u_{n+1}^{m+1} - 2u_n^{m+1} + u_{n-1}^{m+1}\big) \geq b_n^m. \tag{41}$$

This means that, once the solution at time level m is known, the RHS b_n^m is explicit, and we only need to solve for the unknown values at time level $m+1$. The inequality constraint is also discretized as

$$u_n^{m+1} \geq g_n^{m+1}. \tag{42}$$

Combining (41) and (42), the complementarity condition becomes

$$\Big(u_n^{m+1} - \alpha\theta\big(u_{n+1}^{m+1} - 2u_n^{m+1} + u_{n-1}^{m+1}\big) - b_n^m\Big)\big(u_n^{m+1} - g_n^{m+1}\big) = 0. \tag{43}$$

It is therefore natural to write the problem at each time level in matrix form as

$$\mathbf{C}\mathbf{u}^{m+1} \geq \mathbf{b}^m, \qquad \mathbf{u}^{m+1} \geq \mathbf{g}^{m+1}, \qquad (\mathbf{u}^{m+1} - \mathbf{g}^{m+1})^\top(\mathbf{C}\mathbf{u}^{m+1} - \mathbf{b}^m) = 0, \tag{44}$$

where

$$\mathbf{C} = \begin{pmatrix} 1+2\alpha\theta & -\alpha\theta & 0 & \cdots & 0 \\ -\alpha\theta & 1+2\alpha\theta & -\alpha\theta & \cdots & 0 \\ 0 & -\alpha\theta & 1+2\alpha\theta & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & -\alpha\theta \\ 0 & \cdots & 0 & -\alpha\theta & 1+2\alpha\theta \end{pmatrix}, \tag{45}$$

and

$$\mathbf{u}^{m+1} = \begin{pmatrix} u^{m+1}_{-N^-+1} \\ \vdots \\ u^{m+1}_{N^+-1} \end{pmatrix}, \qquad \mathbf{b}^m = \begin{pmatrix} b^m_{-N^-+1} \\ \vdots \\ b^m_{N^+-1} \end{pmatrix}, \qquad \mathbf{g}^{m+1} = \begin{pmatrix} g^{m+1}_{-N^-+1} \\ \vdots \\ g^{m+1}_{N^+-1} \end{pmatrix}. \tag{46}$$

IC & BC The initial condition is inherited directly from the transformed payoff:

$$u^0_n = g^0_n. \tag{47}$$

In addition, the boundary conditions are imposed at the truncated computational boundaries. Thus, for the put one uses

$$u^m_{N^+} = g^m_{N^+}, \tag{48}$$

while for the call one imposes

$$u^m_{-N^-} = g^m_{-N^-}. \tag{49}$$

PSOR Method Following the framework in Algorithm 1, the application follows the steps as follows.

1. Mesh construction and initialization. Discretize the truncated (x,τ)-domain by $x_n = n\,\delta_x,\ \ -N^- \le n \le N^+,\ \ \tau_m = m\,\delta_\tau,\ \ m = 0,1,\dots,M$, and define the interior nodes $n = -N^- + 1, \dots, N^+ - 1$. Also set the mesh ratio (39).

2. Impose the BC & IC. At each time level, the boundary values follow directly from the mesh truncation: $u^m_{-N^-} = g^m_{-N^-},\ \ u^m_{N^+} = g^m_{N^+}$, and the initial condition (47).

3. Form the constraint vector at the new time level. For each $m = 0,1,\dots,M-1$, define $\mathbf{g}^{m+1} = \begin{pmatrix} g^{m+1}_{-N^-+1} \\ \vdots \\ g^{m+1}_{N^+-1} \end{pmatrix}$.

4. **Construct the RHS.** Define $\mathbf{b}^m = \begin{pmatrix} b^m_{-N^-+1} \\ \vdots \\ b^m_{N^+-1} \end{pmatrix}$,

   - Leftmost: $b^m_{-N^-+1} = v^m_{-N^-+1} + \alpha(1-\theta)\big(v^m_{-N^-+2} - 2v^m_{-N^-+1} + g^m_{-N^-}\big) + \alpha\theta\, g^{m+1}_{-N^-}$,
   - Interior entries: $b^m_n = v^m_n + \alpha(1-\theta)\big(v^m_{n+1} - 2v^m_n + v^m_{n-1}\big),\ \ -N^- + 2 \le n \le N^+ - 2$,
   - Rightmost: $b^m_{N^+-1} = v^m_{N^+-1} + \alpha(1-\theta)\big(g^m_{N^+} - 2v^m_{N^+-1} + v^m_{N^+-2}\big) + \alpha\theta\, g^{m+1}_{N^+}$.

5. Initialize the PSOR iteration. Let $\mathbf{V}^0 = \max(\mathbf{v}^m, \mathbf{g}^{m+1})$.

6. **Update interior nodes.**

   - Leftmost: Compute

   $$U^{j+1}_{-N^-+1} = \frac{b^m_{-N^-+1} + \alpha\theta V^j_{-N^-+2}}{1+2\alpha\theta}$$

   Update

   $$V^{j+1}_{-N^-+1} = \max\left(g^{m+1}_{-N^-+1},\ V^j_{-N^-+1} + \omega\big(U^{j+1}_{-N^-+1} - V^j_{-N^-+1}\big)\right).$$

- Interior entries: For $n = -N^- + 2, \ldots, N^+ - 2$, compute

$$U_n^{j+1} = \frac{b_n^m + \alpha\theta V_{n-1}^{j+1} + \alpha\theta V_{n+1}^{j}}{1 + 2\alpha\theta},$$

  and update

$$V_n^{j+1} = \max\left(g_n^{m+1},\ V_n^j + \omega\big(U_n^{j+1} - V_n^j\big)\right).$$

- Rightmost: Compute

$$U_{N^+-1}^{j+1} = \frac{b_{N^+-1}^m + \alpha\theta V_{N^+-2}^{j+1}}{1 + 2\alpha\theta},$$

  and then update

$$V_{N^+-1}^{j+1} = \max\left(g_{N^+-1}^{m+1},\ V_{N^+-1}^j + \omega\big(U_{N^+-1}^{j+1} - V_{N^+-1}^j\big)\right).$$

7. Check convergence. Stop the inner iteration when $\sum_i \big(V_i^{j+1} - V_i^j\big)^2 \le \varepsilon^2$.
8. Advance to the next time level. Once the iteration has converged, set $\mathbf{v}^{m+1} = \mathbf{V}^{j+1}$, and continue to the next time step until all time levels have been completed.

Numerical Example: American Put The algorithm is applied to a American put option, following the same parameter setting as in the book. We consider an American put option with

- strike price: $E = 10$
- risk-free interest rate: $r = 0.10$
- volatility: $\sigma = 0.40$.

The transformed problem is solved on the truncated spatial domain $x \in [-6, 6]$, using a uniform mesh with $\delta_x = 0.01$, with a mesh ratio $\alpha = 1$. For the time-stepping scheme, we use the CN method ($\theta = \frac{1}{2}$.) Two maturities are considered, with $T = 0.25$, $T = 0.50$, corresponding to three months and six months, respectively. The numerical values are reported at the initial time for the asset prices $S = [0, 2, 4, 6, 8, 10, 12, 14, 16]$. Additionally, the relaxation parameter ω is selected empirically. We test a range of candidate values of ω and record the number of PSOR iterations required for convergence under the same tolerance. The value giving the smallest iteration count is then chosen as the optimal relaxation parameter for the computation. In this example, $\omega \approx 1.2$ gives the fastest convergence. The results are recorded in Table 3, which can be compared with Figure 5.

Table 3: American put values obtained by the PSOR method

Asset Price S	Payoff Value	3 Months American	6 Months American
0.00	10.0000	10.0000	10.0000
2.00	8.0000	8.0000	8.0000
4.00	6.0000	5.9999	5.9999
6.00	4.0000	4.0000	4.0000
8.00	2.0000	2.0202	2.0953
10.00	0.0000	0.6920	0.9217
12.00	0.0000	0.1712	0.3624
14.00	0.0000	0.0332	0.1321
16.00	0.0000	0.0055	0.0460

After solving the transformed American put problem, the free boundary is identified *a posteriori* from the computed numerical solution in the original asset-price variable. For each transformed time level τ_n, the numerical solution $u(x, \tau_n)$ is first mapped back to the original option value $V(S, t_n)$, where $S = Ee^x$ and t_n is the corresponding calendar time. We then define the premium

$$P(S, t_n) := V(S, t_n) - \max(E - S, 0). \tag{50}$$

Here, $\max(E - S, 0)$ is the intrinsic value of the American put. Therefore, the premium measures the excess of the option value over its immediate-exercise payoff. In theory, the American put satisfies

Asset Price	Payoff Value	3 months		6 months	
		Amer.	Euro.	Amer.	Euro.
0.00	10.0000	10.0000	9.7531	10.0000	9.5123
2.00	8.0000	8.0000	7.7531	8.0000	7.5123
4.00	6.0000	6.0000	5.7531	6.0000	5.5128
6.00	4.0000	4.0000	3.7569	4.0000	3.5583
8.00	2.0000	2.0200	1.9024	2.0951	1.9181
10.00	0.0000	0.6913	0.6694	0.9211	0.8703
12.00	0.0000	0.1711	0.1675	0.3622	0.3477
14.00	0.0000	0.0332	0.0326	0.1320	0.1279
16.00	0.0000	0.0055	0.0054	0.0460	0.0448

Figure 21.5: Crank–Nicolson solution for an American put with $E = 10$, $r = 0.1$, $\sigma = 0.4$ and with expiry times of three and six months.

Figure 5: Book Example of PSOR method for American Put Options

$V(S,t) \geq \max(E - S, 0)$, so that $P(S,t) \geq 0$. The free boundary $S_f(t)$ is characterized by the transition between the exercise region and the continuation region. More precisely, let S_0 denote the last point in the exercise region and S_1 the first point in the continuation region, with corresponding premiums P_0 and P_1. Then the free boundary is approximated by the zero crossing of the premium:

$$S_f(t) \approx S_0 - \frac{P_0}{P_1 - P_0}(S_1 - S_0). \tag{51}$$

The detected free boundary at the initial time is approximately $S_f(0) \approx 7.558$ for the 3-month case and $S_f(0) \approx 7.118$ for the 6-month case. This makes sense since a longer-maturity American put has more remaining time value, it is less optimal to exercise early, so the critical exercise threshold is lower for the 6-month option. Near maturity, the detected free boundary is approximately $S_f(T) \approx 9.608$ for both cases, which is close to the strike price. The detection result is shown in Figure 6.

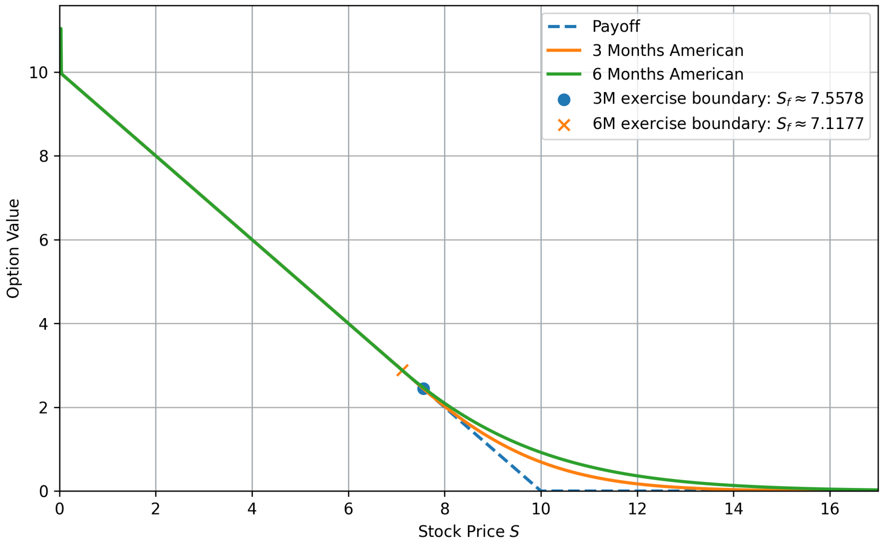

Figure 6: Free BD Detection for American Put

2.2 Transformation 2: VI

2.2.1 Obstacle Problem

We still start from the simple obstacle problem, and reformulate into a variational inequality. The key idea is to characterize the solution through an integral inequality over a suitable set of admissible functions. Let K denote the set of all functions $v(x)$ such that

$$v(-1) = v(1) = 0, \quad v(x) \geq f(x) \text{ for } -1 \leq x \leq 1, \tag{52}$$

and v is continuous with piecewise continuous derivative. The functions in K are called test functions, and K forms the admissible space. For any $v \in K$, we have $v - f \geq 0$. Moreover, from (7), we know that $-u'' \geq 0$. Multiplying these two nonnegative quantities yields

$$-u''(v - f) \geq 0. \tag{53}$$

Integrating over the domain gives

$$\int_{-1}^{1} -u''(v - f)\, dx \geq 0. \tag{54}$$

On the other hand, by the complementarity condition satisfied by the solution u, we also have

$$\int_{-1}^{1} -u''(u - f)\, dx = 0. \tag{55}$$

Then, (54) subtracts (55) eliminates the obstacle f and leads to

$$\int_{-1}^{1} -u''(v - u)\, dx \geq 0. \tag{56}$$

To obtain a weak formulation, we integrate by parts:

$$\int_{-1}^{1} -u''(v - u)\, dx = [-u'(v - u)]_{-1}^{1} + \int_{-1}^{1} u'(v - u)'\, dx. \tag{57}$$

Since both u and v satisfy the boundary conditions, we have $v - u = 0$ at $x = \pm 1$, and thus the boundary term vanishes. Hence,

$$\int_{-1}^{1} u'(v - u)'\, dx \geq 0. \tag{58}$$

This motivates the variational inequality formulation of the obstacle problem. We need to find $u \in K$ such that

$$\int_{-1}^{1} u'(v - u)'\, dx \geq 0 \quad \text{for every } v \in K. \tag{59}$$

Since this formulation only involves first derivatives, which makes it particularly attractive for numerical approximation methods such as FEM.

2.2.2 American options

Given the PDE is parabolic, the VI arises from this problem is called a parabolic VI (the obstacle problem's VI corresponds to a elliptic VI). We first need the LCP form in (30), with the IC and BC: $u(x, 0) = g(x, 0)$, and $u(-x^-, \tau) = g(-x^-, \tau); u(x^+, \tau) = g(x^+, \tau)$. We also need the conditions that $u(x, \tau)$ and $\frac{\partial u}{\partial x}(x, \tau)$ are continuous. Following the pattern of the obstacle problem, let K be the space of admissible test functions $\phi(x, \tau)$ such that:

- ϕ and $\partial\phi/\partial\tau$ are continuous, and $\partial\phi/\partial x$ is piecewise continuous;
- $\phi(x, \tau) \geq g(x, \tau)$ for all x, τ;
- $\phi(x^+, \tau) = g(x^+, \tau) = 0$ and $\phi(-x^-, \tau) = g(-x^-, \tau)$;
- $\phi(x, 0) = g(x, 0)$.

The solution $u(x, \tau)$ belongs to K. For any $\phi \in K$, since $\phi \geq g$, we have

$$\left(\frac{\partial u}{\partial \tau} - \frac{\partial^2 u}{\partial x^2}\right)(\phi - g) \geq 0. \tag{60}$$

Integrating over space yields

$$\int_{-x^-}^{x^+} \left(\frac{\partial u}{\partial \tau} - \frac{\partial^2 u}{\partial x^2}\right)(\phi - g)\, dx \geq 0. \tag{61}$$

Similarly, using the complementarity condition,

$$\int_{-x^-}^{x^+} \left(\frac{\partial u}{\partial \tau} - \frac{\partial^2 u}{\partial x^2}\right)(u - g)\, dx = 0. \tag{62}$$

Subtracting eliminates g:

$$\int_{-x^-}^{x^+} \left(\frac{\partial u}{\partial \tau} - \frac{\partial^2 u}{\partial x^2}\right)(\phi - u)\, dx \geq 0. \tag{63}$$

Applying integration by parts, we obtain

$$\int_{-x^-}^{x^+} \frac{\partial u}{\partial \tau}(\phi - u) + \frac{\partial u}{\partial x}\left(\frac{\partial \phi}{\partial x} - \frac{\partial u}{\partial x}\right) dx \geq 0, \tag{64}$$

since boundary terms vanish due to the admissibility conditions. For the VI formulation, we need to find $u(x, \tau) \in K$ such that, for all $\phi \in K$ and all $0 \leq \tau \leq \frac{1}{2}\sigma^2 T$,

$$\int_{-x^-}^{x^+} \frac{\partial u}{\partial \tau}(\phi - u) + \frac{\partial u}{\partial x}\left(\frac{\partial \phi}{\partial x} - \frac{\partial u}{\partial x}\right) dx \geq 0. \tag{65}$$

An equivalent global formulation is

$$\int_0^{\frac{1}{2}\sigma^2 T} \int_{-x^-}^{x^+} \frac{\partial u}{\partial \tau}(\phi - u) + \frac{\partial u}{\partial x}\left(\frac{\partial \phi}{\partial x} - \frac{\partial u}{\partial x}\right) dx\, d\tau \geq 0. \tag{66}$$

It can be shown that the solution of $u(x, \tau) \in K$ of the VI ((65) or (66)) is a solution of LCP (30), and the free boundary problems (1)–(5).

2.3 Finite-Element Methods

FEM suits the free boundary problems in VI form, but this section will show that it ultimately leads to the same numerical problem as FDM.

2.3.1 Obstacle Problem

Finite Element Formulation The variational inequality (59) is infinite-dimensional, since the admissible set K contains all continuous functions satisfying the obstacle and boundary constraints. To compute a numerical solution, we replace K by a finite-dimensional subspace K_n and approximate the solution by a linear combination of basis functions. Specifically, we divide the interval $(-1, 1)$ into a mesh $-1 = x_0 < x_1 < \cdots < x_n = 1$, and introduce basis functions $\{\psi_1(x), \psi_2(x), \ldots, \psi_n(x)\}$ associated with the nodes. These are often called shape functions. The approximate solution is written as

$$u_h(x) = \sum_{i=1}^{n} u_i \psi_i(x), \tag{67}$$

where the coefficients u_i are unknown nodal values to be determined. Similarly, every admissible test function in the discrete space is represented by

$$v_h(x) = \sum_{j=1}^{n} v_j \psi_j(x), \tag{68}$$

where the coefficients v_j satisfy the obstacle constraint

$$v_h(x) \geq f(x), \quad -1 \leq x \leq 1. \tag{69}$$

The discrete solution must also remain above the obstacle, so we require

$$u_h(x) \geq f(x), \quad -1 \leq x \leq 1. \tag{70}$$

Substituting u_h and v_h into the variational inequality gives

$$\int_{-1}^{1} u_h'(v_h - u_h)'\, dx \geq 0 \quad \text{for all admissible } v_h. \tag{71}$$

Using

$$u_h'(x) = \sum_{i=1}^{n} u_i \psi_i'(x), \qquad (v_h - u_h)' = \sum_{j=1}^{n} (v_j - u_j)\psi_j'(x), \tag{72}$$

we obtain

$$\int_{-1}^{1} \left(\sum_{i=1}^{n} u_i \psi_i'(x) \right) \left(\sum_{j=1}^{n} (v_j - u_j)\psi_j'(x) \right) dx \geq 0. \tag{73}$$

Expanding the sums leads to

$$\sum_{i=1}^{n} \sum_{j=1}^{n} u_i (v_j - u_j) \int_{-1}^{1} \psi_i'(x)\psi_j'(x)\, dx \geq 0. \tag{74}$$

This motivates the stiffness matrix $\mathbf{A} = (A_{ij})$, defined by

$$A_{ij} = \int_{-1}^{1} \psi_i'(x)\psi_j'(x)\, dx. \tag{75}$$

Hence the finite element variational inequality becomes

$$\sum_{i=1}^{n} \sum_{j=1}^{n} A_{ij} u_i (v_j - u_j) \geq 0. \tag{76}$$

If we denote the coefficient vectors by $\mathbf{u} = (u_1, u_2, \ldots, u_n)^T$ and $\mathbf{v} = (v_1, v_2, \ldots, v_n)^T$, then the discrete problem can be written compactly as

$$\mathbf{u}^T \mathbf{A}(\mathbf{v} - \mathbf{u}) \geq 0. \tag{77}$$

Therefore, the finite element approximation of the obstacle problem is: find $\mathbf{u}$ such that the reconstructed function $u_h(x)$ satisfies (70) and (77) for every admissible vector $\mathbf{v}$ whose corresponding function $v_h(x)$ also satisfies (69).

Implementation: Choice of Basis Functions To implement the finite element method, we now specify the basis functions used in the discrete space. We partition the interval $(-1, 1)$ into equally spaced nodes $x_i = i\delta_x,\ \ i = -N, -N+1, \ldots, N$, where $\delta_x = \frac{1}{N}$. Hence $x_{-N} = -1$, $x_0 = 0$, and $x_N = 1$. We take $n = 2N + 1$ interior unknowns corresponding to the interior nodes. Each finite element is the interval $D_i = [x_{i-1}, x_{i+1}]$, centered at node x_i, shown in FIgure 7.

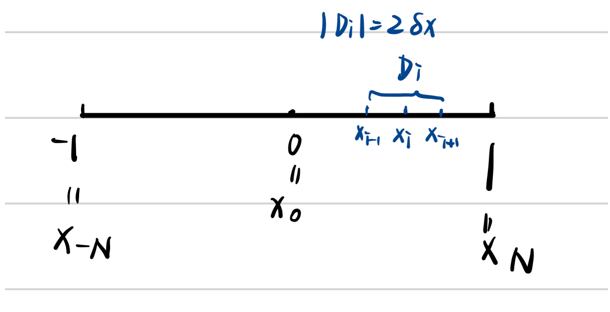

Figure 7: The interval $(-1, 1)$ divided into finite elements.

We choose piecewise linear basis functions $\psi_i(x)$, also called hat functions. Each basis function is supported only on D_i, equals one at its own node, and vanishes at neighboring nodes:

$$\psi_i(x_{i-1}) = 0, \qquad \psi_i(x_i) = 1, \qquad \psi_i(x_{i+1}) = 0. \tag{78}$$

Its explicit form is

$$\psi_i(x) = \begin{cases} 0, & x < x_{i-1}, \\ \dfrac{x - x_{i-1}}{\delta_x}, & x_{i-1} \leq x \leq x_i, \\ \dfrac{x_{i+1} - x}{\delta_x}, & x_i \leq x \leq x_{i+1}, \\ 0, & x > x_{i+1}. \end{cases} \tag{79}$$

Therefore,

$$\psi_i'(x) = \begin{cases} 0, & x < x_{i-1}, \\ \dfrac{1}{\delta_x}, & x_{i-1} < x < x_i, \\ -\dfrac{1}{\delta_x}, & x_i < x < x_{i+1}, \\ 0, & x > x_{i+1}. \end{cases} \tag{80}$$

Using these basis functions, we can write any discrete function $u_h(x_i) = u_i$, $v_h(x_i) = v_i$, and $f_i = f(x_i)$ ($f(x) = \sum_j f_j \psi_j(x)$), given that $\psi_j(x_i) = 1$ only for $i = j$. Then, the stiffness matrix entries (75) are

$$A_{ij} = \frac{1}{\delta_x} \begin{cases} 2, & j = i, \\ -1, & j = i \pm 1, \\ 0, & \text{otherwise}, \end{cases} \tag{81}$$

given that the neighboring basis functions overlap only locally. Hence the matrix takes the tridiagonal form

$$\mathbf{A} = \frac{1}{\delta_x} \begin{pmatrix} 2 & -1 & 0 & \cdots & 0 \\ -1 & 2 & -1 & \ddots & \vdots \\ 0 & -1 & 2 & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & -1 \\ 0 & \cdots & 0 & -1 & 2 \end{pmatrix}. \tag{82}$$

If we define $\mathbf{A} = (\delta_x)^{-1}\mathbf{B}$, then,

$$\mathbf{B} = \begin{pmatrix} 2 & -1 & 0 & \cdots & 0 \\ -1 & 2 & -1 & \ddots & \vdots \\ 0 & -1 & 2 & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & -1 \\ 0 & \cdots & 0 & -1 & 2 \end{pmatrix}, \tag{83}$$

which is exactly the same as (13) in FDM scheme. Thus, the finite element formulation is to find $\mathbf{u} \geq \mathbf{f}$ such that

$$\mathbf{u}^\top \mathbf{B}(\mathbf{v} - \mathbf{u}) \geq 0, \qquad \forall\, \mathbf{v} \geq \mathbf{f}. \tag{84}$$

Proof of Equivalence We now show that the finite element VI (84) is equivalent to the finite difference LCP form (12).

Part I: FEM $\Rightarrow$ LCP Assume $\mathbf{u}$ satisfies (84). Since $\mathbf{B}$ is symmetric,

$$\mathbf{u}^\top \mathbf{B}(\mathbf{v} - \mathbf{u}) = (\mathbf{v} - \mathbf{u})^\top \mathbf{B}\mathbf{u}. \tag{85}$$

Therefore,

$$(\mathbf{v} - \mathbf{u})^\top \mathbf{B}\mathbf{u} \geq 0, \qquad \forall\, \mathbf{v} \geq \mathbf{f}. \tag{86}$$

We now verify the three complementarity conditions.

1. The admissibility constraint already gives

$$\mathbf{u} \geq \mathbf{f}. \tag{87}$$

2. Then, we prove $\mathbf{Bu} \geq 0$. Suppose some component $(\mathbf{Bu})_i < 0$. Choose v_i arbitrarily large while keeping all other components admissible. Then the inner product in (86) becomes negative, contradicting the inequality. Hence,

$$\mathbf{Bu} \geq 0. \tag{88}$$

3. Finally, choose $\mathbf{v} = \mathbf{f}$ in (86). Since $\mathbf{f} \geq \mathbf{f}$ is admissible,

$$(\mathbf{f} - \mathbf{u})^\top \mathbf{B}\mathbf{u} \geq 0. \tag{89}$$

Multiplying by -1 gives

$$(\mathbf{u} - \mathbf{f})^\top \mathbf{B}\mathbf{u} \leq 0. \tag{90}$$

But both vectors $\mathbf{u} - \mathbf{f}$ and $\mathbf{B}\mathbf{u}$ are componentwise nonnegative, so their inner product must also satisfy

$$(\mathbf{u} - \mathbf{f})^\top \mathbf{B}\mathbf{u} \geq 0. \tag{91}$$

Hence the only possibility is

$$(\mathbf{u} - \mathbf{f})^\top \mathbf{B}\mathbf{u} = 0. \tag{92}$$

Thus, from (87), (88), and (91), we prove that (12) holds.

Part II: LCP $\Rightarrow$ FEM Now assume $\mathbf{u}$ satisfies (12). Let any admissible vector $\mathbf{v} \geq \mathbf{f}$ be given. Since $\mathbf{v} - \mathbf{f} \geq 0$ and $\mathbf{B}\mathbf{u} \geq 0$, we have

$$(\mathbf{v} - \mathbf{f})^\top \mathbf{B}\mathbf{u} \geq 0. \tag{93}$$

Using the complementarity condition,

$$(\mathbf{u} - \mathbf{f})^\top \mathbf{B}\mathbf{u} = 0. \tag{94}$$

Subtracting these two relations yields

$$(\mathbf{v} - \mathbf{u})^\top \mathbf{B}\mathbf{u} \geq 0. \tag{95}$$

Since $\mathbf{B}$ is symmetric,

$$(\mathbf{v} - \mathbf{u})^\top \mathbf{B}\mathbf{u} = \mathbf{u}^\top \mathbf{B}(\mathbf{v} - \mathbf{u}). \tag{96}$$

Therefore,

$$\mathbf{u}^\top \mathbf{B}(\mathbf{v} - \mathbf{u}) \geq 0, \qquad \forall\, \mathbf{v} \geq \mathbf{f}, \tag{97}$$

which is exactly the finite element variational inequality (84).

2.3.2 American Options

Finite Element Formulation The variational inequality (65) is infinite-dimensional, since the admissible set K contains all space-time functions satisfying the payoff, boundary, and initial constraints. To compute a numerical solution, we replace K by a finite-dimensional subspace and approximate the solution using basis functions in space. Similarly, we divide the spatial interval $(-x^-, x^+)$ into a mesh $-x^- = x_0 < x_1 < \cdots < x_n = x^+$, and introduce time-independent basis functions $\{\psi_1(x), \psi_2(x), \ldots, \psi_n(x)\}$. Since the American option problem is time-dependent, the coefficients of the basis functions depend on τ. Thus the approximate solution is written as

$$u_h(x,\tau) = \sum_{i=1}^{n} u_i(\tau)\psi_i(x), \tag{98}$$

where $u_i(\tau)$ are unknown time-dependent nodal values. Similarly, every admissible test function is represented by

$$\phi_h(x,\tau) = \sum_{j=1}^{n} \phi_j(\tau)\psi_j(x), \tag{99}$$

where the coefficients $\phi_j(\tau)$ satisfy the payoff constraint

$$\phi_h(x,\tau) \geq g(x,\tau), \quad -x^- \leq x \leq x^+. \tag{100}$$

The discrete solution must also remain above the payoff, so we require

$$u_h(x,\tau) \geq g(x,\tau), \quad -x^- \leq x \leq x^+. \tag{101}$$

Substituting u_h and ϕ_h into the variational inequality (65) gives

$$\int_{-x^-}^{x^+} \frac{\partial u_h}{\partial \tau}(\phi_h - u_h) + \frac{\partial u_h}{\partial x}\left(\frac{\partial \phi_h}{\partial x} - \frac{\partial u_h}{\partial x}\right) dx \geq 0. \tag{102}$$

Using

$$\frac{\partial u_h}{\partial \tau} = \sum_{i=1}^{n} \dot{u}_i(\tau)\psi_i(x), \qquad \frac{\partial u_h}{\partial x} = \sum_{i=1}^{n} u_i(\tau)\psi_i'(x), \tag{103}$$

and

$$\phi_h - u_h = \sum_{j=1}^{n} \big(\phi_j(\tau) - u_j(\tau)\big)\psi_j(x), \tag{104}$$

we obtain

$$\sum_{i=1}^{n}\sum_{j=1}^{n} M_{ij}\dot{u}_i(\tau)(\phi_j - u_j) + \sum_{i=1}^{n}\sum_{j=1}^{n} K_{ij}u_i(\tau)(\phi_j - u_j) \geq 0, \tag{105}$$

where the mass matrix $\mathbf{M} = (M_{ij})$ and stiffness matrix $\mathbf{K} = (K_{ij})$ are defined by

$$M_{ij} = \int_{-x^-}^{x^+} \psi_i(x)\psi_j(x)\, dx, \qquad K_{ij} = \int_{-x^-}^{x^+} \psi_i'(x)\psi_j'(x)\, dx. \tag{106}$$

If we denote the coefficient vectors by

$$\mathbf{u}(\tau) = \big(u_1(\tau), u_2(\tau), \ldots, u_n(\tau)\big)^T, \qquad \boldsymbol{\phi}(\tau) = \big(\phi_1(\tau), \phi_2(\tau), \ldots, \phi_n(\tau)\big)^T, \tag{107}$$

then the discrete problem can be written compactly as

$$(\boldsymbol{\phi} - \mathbf{u})^T (\mathbf{M}\dot{\mathbf{u}} + \mathbf{K}\mathbf{u}) \geq 0. \tag{108}$$

Therefore, the finite element approximation of the American option problem is: find $\mathbf{u}(\tau)$ such that the reconstructed solution $u_h(x,\tau)$ satisfies (101) and (108) for every admissible vector $\boldsymbol{\phi}(\tau)$ whose corresponding function $\phi_h(x,\tau)$ also satisfies (100).

Next, to discretize the time variable, we divide the interval $0 \leq \tau \leq \frac{1}{2}\sigma^2 T$ into steps of length $\delta\tau$, and define $\tau_m = m\delta\tau$, $\mathbf{u}^m = \mathbf{u}(\tau_m)$. Using the forward difference approximation, the time derivative is written as

$$\dot{\mathbf{u}}(\tau_m) \approx \frac{\mathbf{u}^{m+1} - \mathbf{u}^m}{\delta\tau}. \tag{109}$$

To treat the spatial operator, we apply the standard θ-scheme as above:

$$\mathbf{K}\mathbf{u} \approx \theta\mathbf{K}\mathbf{u}^{m+1} + (1-\theta)\mathbf{K}\mathbf{u}^m. \tag{110}$$

Substituting these approximations into (108), and evaluating the test vector at the new time level, $\boldsymbol{\phi}^{m+1} \geq \mathbf{f}^{m+1}$, we obtain the fully discrete variational inequality

$$(\boldsymbol{\phi}^{m+1} - \mathbf{u}^{m+1})^T \left[\mathbf{M}\frac{\mathbf{u}^{m+1} - \mathbf{u}^m}{\delta\tau} + \mathbf{K}\big(\theta\mathbf{u}^{m+1} + (1-\theta)\mathbf{u}^m\big)\right] \geq 0. \tag{111}$$

Multiplying by $\delta\tau$ gives the equivalent form

$$(\boldsymbol{\phi}^{m+1} - \mathbf{u}^{m+1})^T \left[\mathbf{M}(\mathbf{u}^{m+1} - \mathbf{u}^m) + \delta\tau\mathbf{K}\big(\theta\mathbf{u}^{m+1} + (1-\theta)\mathbf{u}^m\big)\right] \geq 0. \tag{112}$$

Therefore, at each time step, the unknown vector $\mathbf{u}^{m+1}$ is determined by solving a discrete complementarity-type variational inequality subject to $\mathbf{u}^{m+1} \geq \mathbf{f}^{m+1}$. This is the finite element time-marching formulation for the American option problem.

Apply Similar Basis Function To obtain a practical numerical scheme, we now choose the same piecewise linear hat functions used in the obstacle problem. Divide the interval $[-x^-, x^+]$ into equally spaced nodes $x_i = -x^- + i\delta x, \quad i = 0, 1, \ldots, n$, where $x_0 = -x^-$ and $x_n = x^+$. We define the basis functions $\psi_i(x)$ by

$$\psi_i(x) = \frac{1}{\delta x}\begin{cases} 0, & x < x_{i-1}, \\ x - x_{i-1}, & x_{i-1} \le x < x_i, \\ x_{i+1} - x, & x_i \le x < x_{i+1}, \\ 0, & x \ge x_{i+1}. \end{cases} \tag{113}$$

As before, the nodal coefficients satisfy $u_i(\tau) = u(x_i, \tau)$, $\phi_i(\tau) = \phi(x_i, \tau)$, and $g_i(\tau) = g(x_i, \tau)$. Hence the payoff constraints reduce to

$$\mathbf{u}^m \ge \mathbf{g}^m, \qquad \boldsymbol{\phi}^m \ge \mathbf{g}^m, \tag{114}$$

where $\mathbf{g}^m = \big(g_1(\tau_m), g_2(\tau_m), \ldots, g_n(\tau_m)\big)^T$. Then, the initial condition becomes $\mathbf{u}^0 = \mathbf{g}^0$. With these basis functions, both matrices $\mathbf{M}$ and $\mathbf{K}$ are sparse tridiagonal matrices. In particular,

$$K_{ij} = \frac{1}{\delta x}\begin{cases} 2, & i = j, \\ -1, & i = j \pm 1, \\ 0, & \text{otherwise}. \end{cases} \tag{115}$$

This is the same as the format in (82), thus $\mathbf{B} = \delta_x \mathbf{K}$ is the same as (83). Then, we examine the mass matrix $\mathbf{M}$. To simplify the numerical scheme, we apply the Rectangle Rule:

$$\int_a^b f(x)\,dx \approx (b-a)\sum f(x), \tag{116}$$

where each basis function spans two adjacent subintervals, so the effective local width is $2\delta x$, same in Figure 7. Applying this rule to $f = \psi_i(x)\psi_j(x)$ gives

$$M_{ij} = \int_{-x^-}^{x^+} \psi_i(x)\psi_j(x)\,dx \approx 2\delta x \sum_{k=1}^{n} \psi_i(x_k)\psi_j(x_k). \tag{117}$$

Since each hat basis function ψ_i has local support only on $[x_{i-1}, x_{i+1}]$, the product $\psi_i(x)\psi_j(x)$ is nonzero only when the supports overlap. Thus, $M_{ij} = 0, \quad i \ne j$, and $M_{ii} = 2\delta x$. Since the common scalar factor does not affect the variational inequality after rescaling, we absorb the factor 2 into the formulation and write

$$\mathbf{M} \approx \delta x\, \mathbf{I}. \tag{118}$$

Hence the mass matrix becomes diagonal. Substituting these approximations into the fully discrete variational inequality gives

$$(\boldsymbol{\phi}^{m+1} - \mathbf{u}^{m+1})^T \left[\mathbf{u}^{m+1} + \frac{\delta\tau}{\delta x}\theta \mathbf{B}\mathbf{u}^{m+1} - \mathbf{u}^m + \frac{\delta\tau}{\delta x}(1-\theta)\mathbf{B}\mathbf{u}^m\right] \ge 0. \tag{119}$$

Using the mesh ratio α in (39), we obtain

$$(\boldsymbol{\phi}^{m+1} - \mathbf{u}^{m+1})^T \left[(\mathbf{I} + \alpha\theta\mathbf{B})\mathbf{u}^{m+1} - \big(\mathbf{I} - \alpha(1-\theta)\mathbf{B}\big)\mathbf{u}^m\right] \ge 0. \tag{120}$$

Define

$$\mathbf{C} = \mathbf{I} + \alpha\theta\mathbf{B}, \tag{121}$$

which shares the same format as (45) and

$$\mathbf{b}^m = \big(\mathbf{I} - \alpha(1-\theta)\mathbf{B}\big)\mathbf{u}^m. \tag{122}$$

Then the finite element scheme becomes

$$(\boldsymbol{\phi}^{m+1} - \mathbf{u}^{m+1})^T(\mathbf{C}\mathbf{u}^{m+1} - \mathbf{b}^m) \ge 0. \tag{123}$$

Therefore, at each time level, the finite element approximation of the American option problem reduces to solving a LCP system subject to $\mathbf{u}^{m+1} \ge \mathbf{g}^{m+1}$, for every admissible test vector satisfying $\boldsymbol{\phi}^{m+1} \ge \mathbf{g}^{m+1}$.

Proof of Equivalence Similarly, we prove that the finite element problem (123) is equivalent to the linear complementarity system (44). The proof is carried out in two directions.

Part I: LCP to FEM From (123), LHS is

$$(\boldsymbol{\phi}^{m+1}-\mathbf{u}^{m+1})^T(\mathbf{C}\mathbf{u}^{m+1}-\mathbf{b}^m)=(\boldsymbol{\phi}^{m+1}-\mathbf{g}^{m+1})^T(\mathbf{C}\mathbf{u}^{m+1}-\mathbf{b}^m)-(\mathbf{u}^{m+1}-\mathbf{g}^{m+1})^T(\mathbf{C}\mathbf{u}^{m+1}-\mathbf{b}^m). \tag{124}$$

Since $\mathbf{u}^{m+1}$ satisfies the linear complementarity system (44). Then

$$\mathbf{C}\mathbf{u}^{m+1}-\mathbf{b}^m\geq 0, \tag{125}$$

and

$$(\mathbf{u}^{m+1}-\mathbf{g}^{m+1})^T(\mathbf{C}\mathbf{u}^{m+1}-\mathbf{b}^m)=0. \tag{126}$$

Hence

$$(\boldsymbol{\phi}^{m+1}-\mathbf{u}^{m+1})^T(\mathbf{C}\mathbf{u}^{m+1}-\mathbf{b}^m)=(\boldsymbol{\phi}^{m+1}-\mathbf{g}^{m+1})^T(\mathbf{C}\mathbf{u}^{m+1}-\mathbf{b}^m). \tag{127}$$

Since every admissible test vector satisfies $\boldsymbol{\phi}^{m+1}\geq\mathbf{g}^{m+1}$, we have

$$(\boldsymbol{\phi}^{m+1}-\mathbf{g}^{m+1})^T(\mathbf{C}\mathbf{u}^{m+1}-\mathbf{b}^m)\geq 0. \tag{128}$$

Therefore,

$$(\boldsymbol{\phi}^{m+1}-\mathbf{u}^{m+1})^T(\mathbf{C}\mathbf{u}^{m+1}-\mathbf{b}^m)\geq 0, \tag{129}$$

which is exactly the finite element variational inequality (123).

Part II: FEM to LCP Now assume that $\mathbf{u}^{m+1}$ satisfies

$$(\boldsymbol{\phi}^{m+1}-\mathbf{u}^{m+1})^T(\mathbf{C}\mathbf{u}^{m+1}-\mathbf{b}^m)\geq 0 \tag{130}$$

for every admissible vector $\boldsymbol{\phi}^{m+1}\geq\mathbf{g}^{m+1}$, together with the constraint $\mathbf{u}^{m+1}\geq\mathbf{g}^{m+1}$. We first prove that

$$\mathbf{C}\mathbf{u}^{m+1}-\mathbf{b}^m\geq 0. \tag{131}$$

Suppose, on the contrary, that some component of $\mathbf{C}\mathbf{u}^{m+1}-\mathbf{b}^m<0$. Choose the corresponding component of $\boldsymbol{\phi}^{m+1}$ arbitrarily large, while keeping all other components admissible and satisfying $\boldsymbol{\phi}^{m+1}\geq\mathbf{g}^{m+1}$. Then the quantity $(\boldsymbol{\phi}^{m+1}-\mathbf{u}^{m+1})^T(\mathbf{C}\mathbf{u}^{m+1}-\mathbf{b}^m)<0$, contradicting (123). Hence

$$\mathbf{C}\mathbf{u}^{m+1}-\mathbf{b}^m\geq 0. \tag{132}$$

Next, since $\mathbf{u}^{m+1}\geq\mathbf{g}^{m+1}$, we obtain $(\mathbf{u}^{m+1}-\mathbf{g}^{m+1})^T(\mathbf{C}\mathbf{u}^{m+1}-\mathbf{b}^m)\geq 0$. On the other hand, choose the admissible test vector $\boldsymbol{\phi}^{m+1}=\mathbf{g}^{m+1}$. and substitute into (123) gives

$$(\mathbf{g}^{m+1}-\mathbf{u}^{m+1})^T(\mathbf{C}\mathbf{u}^{m+1}-\mathbf{b}^m)\geq 0, \tag{133}$$

or equivalently,

$$(\mathbf{u}^{m+1}-\mathbf{g}^{m+1})^T(\mathbf{C}\mathbf{u}^{m+1}-\mathbf{b}^m)\leq 0. \tag{134}$$

Combining the two inequalities yields

$$(\mathbf{u}^{m+1}-\mathbf{g}^{m+1})^T(\mathbf{C}\mathbf{u}^{m+1}-\mathbf{b}^m)=0. \tag{135}$$

Therefore,

$$\mathbf{C}\mathbf{u}^{m+1}\geq\mathbf{b}^m,\qquad \mathbf{u}^{m+1}\geq\mathbf{g}^{m+1},\qquad (\mathbf{u}^{m+1}-\mathbf{g}^{m+1})^T(\mathbf{C}\mathbf{u}^{m+1}-\mathbf{b}^m)=0, \tag{136}$$

which is precisely the linear complementarity system (44). Hence the finite element variational inequality formulation and the finite difference linear complementarity formulation produce the same numerical problem at each time level.

Therefore, The importance of the FEM here is not that it gives a different numerical solution, since it leads to the same LCP system as the FDM. Its real advantage is theoretical: Because the FEM starts from the weak variational inequality, it fits into functional analysis and energy methods. This makes it easier to prove that the numerical solution converges to the true solution as the mesh is refined, and to derive rigorous error bounds. In particular, the convergence of iterative solvers such as the projected SOR method is usually proved more cleanly in the FEM variational setting than in the finite difference LCP form, seen in [1]. *The above analysis is based on Chapter 6, 7, 20, 21, and Appendix D in the book [2].*

References

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